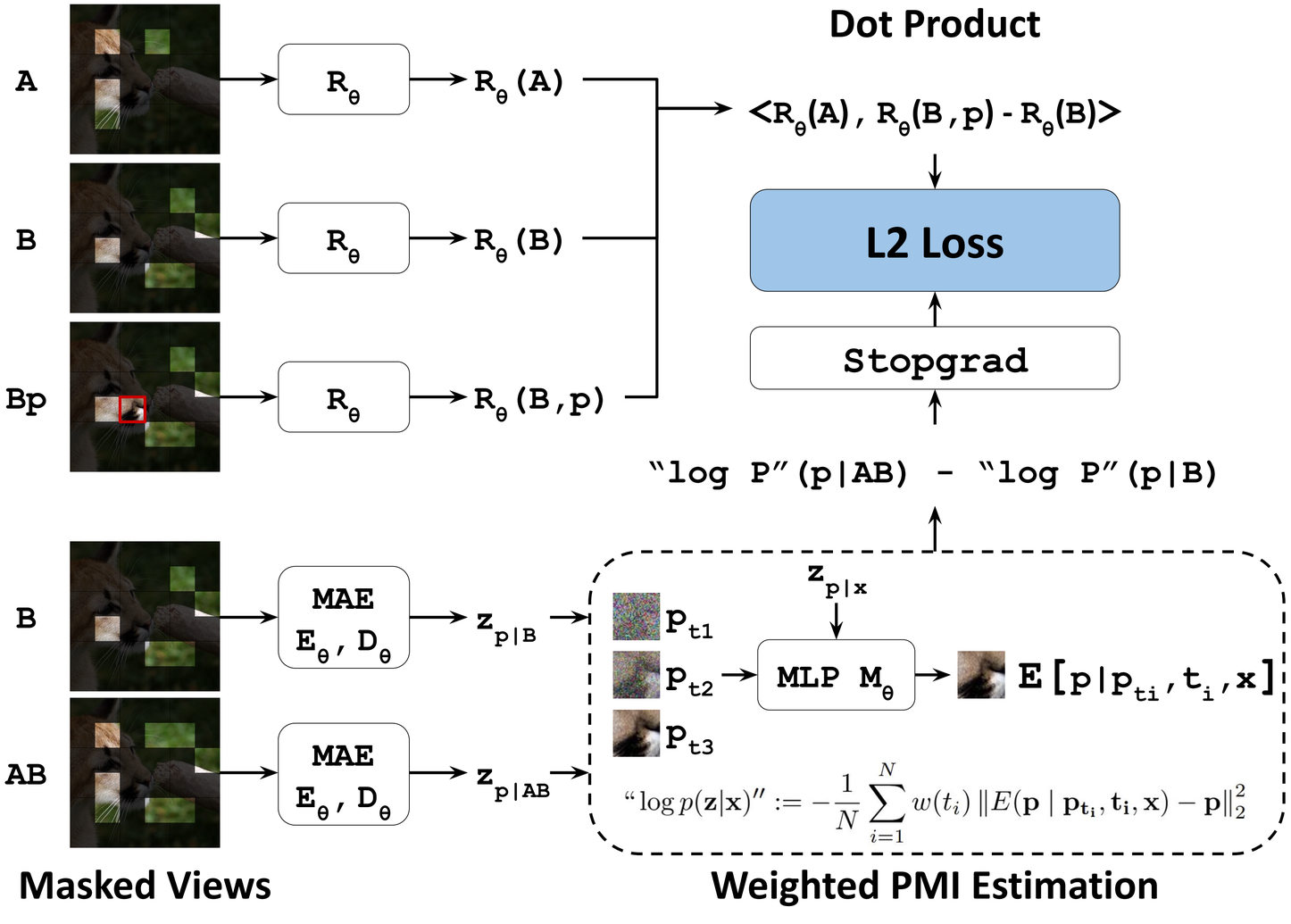
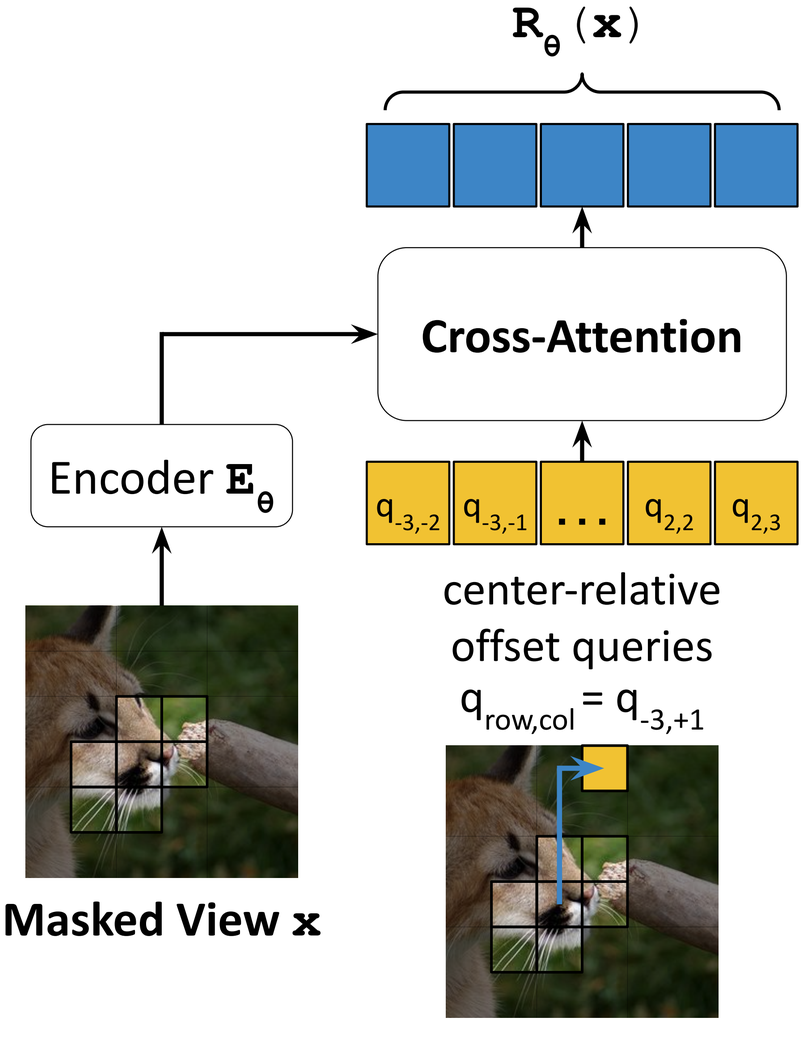

A
B
Bp
R_θ
$R_\theta(A)$
$R_\theta(B)$
$R_\theta(B,p)$
Dot Product
$<R_\theta(A), R_\theta(B,p) - R_\theta(B)>$
L2 Loss
Stopgrad
"log P"(p|AB) - "log P"(p|B)
B
AB
MAE E_θ, D_θ
$z_{p|B}$
$z_{p|AB}$
$z_{p|x}$
p_{t1}
p_{t2}
p_{t3}
MLP M_θ
$E[p|p_{ti}, t_i, x]$
"$\log p(\mathbf{z}|\mathbf{x})$" $:= -\frac{1}{N}\sum_{i=1}^{N} w(t_i)\,\|E(\mathbf{p} \mid \mathbf{p}_{\mathbf{t_i}}, \mathbf{t_i}, \mathbf{x}) - \mathbf{p}\|_2^2$
Masked Views
Weighted PMI Estimation


$R_\theta(x)$
Cross-Attention
Encoder E_θ
$q_{-3,-2}$
$q_{-3,-1}$
...
$q_{2,2}$
$q_{2,3}$
center-relative offset queries
$q_{row,col} = q_{-3,+1}$
Masked View x